

Orion Nichols

Mechanical Engineering Student | www.linkedin.com/in/orionnichols | son4@humboldt.edu | (530)-601-1688

Education Cal-Poly Humboldt /Mechanical Engineering

Graduated - B.S. Spring 2026 - GPA 3.4

Capstone

Autonomous Robotic Pruning System using Bioinspired Locomotion

Extra

Curriculars

Aviation Club - Spring 2024 to Spring 2025

President

- Assisted with installation & setup for the school's new Advanced Aviation Training Device

ASME Chapter - Fall 2025

President

- Managed the transition from startup local chapter to established Mechanical Engineering community on campus

Comap Math Modeling Competition - Spring 2026

- Developed a Mathematical Model to create practical Phone Battery Load Shapes
-

Work &

Research

Experience

STAD Research Group/ Student Research Assistant; www.stadjueunlee.org

Fall 2024 - Present

- Research on thermal necrosis due to drilling operations into bone

Cal-Poly Humboldt Student Learning Center/ Tutor

Fall 2024

- Tutored students on CAD fundamentals in both Solidworks and AutoCAD
- Assisted Students with their Solid Mechanics and Statics Materials homework

JLSky Consultant/ Engineering Design Consultant

June 2024 - August 2024

- Designed HVAC, P&ID systems using AutoCAD for commercial use
- Assisted in the design of the *kababoniator*, an automated kabob making machine

J.C. Chang & Associates/ Mechanical Engineering Intern

April - August 2023

- Designed HVAC layouts for commercial buildings in AutoCAD
- Completed job walks for ongoing projects and project bids
- Collected information to make informed design decisions to assist with project proposals

SunPower R&D/ Mechanical Engineering Intern

May 2020-Oct 2020

- Researched effects of different weather environments on solar module efficiency
- Conducted study on viability of utilizing microinverters for power generation

Starbucks/ Shift Manager

2019 - Feb 2023

- Assistant Store Manager in charge of scheduling, inventory and planning

Citrus Circuits FRC Robotics Team/ Mechanical Fabrication

August 2015 - June 2018

- Operated and trained new members on various heavy machinery including lathes, mills, and CNC Machinery, as well as shop safety, lab usage, hand tools, and CAD software
-

Skills

CAD: SolidWorks, Fusion 360, AutoCAD, Bluebeam REVU, G-Code

Software: Matlab, Python, Java, Microsoft/Google Suite, Mathematica

Machinery: Lathe, mill, CNC routers, 3D printing, Keyence 3D Microscope, dynamometers